



Gaurav Pathak

Master of Computer Applications(AI and IOT)
Computer Science and Engineering
National Institute of Technology, Patna

+91-7505186838
gauravp.mca24@nitp.ac.in
linkedin.com/in/gauravpathak2003
github.com/GauravPathak2023

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MCA (AI & IoT)	National Institute of Technology, Patna	9.03	2024-present
BCA	Graphic Era Hill University, Haldwani	8.19	2021-24
Senior Secondary	CBSE	92.8%	2018-20

PROJECTS

- Project 1: AI-Powered Customer Support Agent** July 2025 - Aug 2025
Tools: Python, LangChain, OpenAI API, Pinecone, Flask
Developed an AI-driven customer support chatbot using LangChain and OpenAI's GPT model to automate responses for a retail platform. Integrated LangChain's conversational memory and Pinecone for vector-based retrieval of 10,000+ support tickets, enabling context-aware responses. Achieved 90% accuracy in resolving common queries, reducing response time by 50%. Deployed the chatbot via Flask, improving customer satisfaction in a simulated environment by 20%.
- Project 2: Movie Recommendation System** June 2025 - July 2025
Tools: Python, pandas, scikit-learn, Surprise, FastAPI, NumPy, matplotlib
Developed a machine learning-based movie recommendation system using collaborative filtering to enhance user experience on a streaming platform. Processed a dataset of 100,000+ user ratings from the MovieLens dataset, applying matrix factorization (SVD) with the Surprise library to generate personalized movie suggestions, achieving a 15% improvement in recommendation accuracy. Deployed the model via FastAPI for real-time predictions and visualized user preference trends using matplotlib, contributing to a 10% increase in simulated user engagement.
- Project 3: Packet Sniffer & Analyzer** March 2025 - May 2025
Tools: Python, Scapy, pandas, matplotlib, Wireshark, SQLite
Developed a network packet sniffer and analyzer using Scapy to capture and analyze real-time network traffic. Processed 100,000+ packets, extracting features like source/destination IPs, protocols, and packet sizes, and stored them in an SQLite database for efficient querying. Implemented anomaly detection using pandas to identify unusual traffic patterns, achieving 92% accuracy in detecting potential intrusions. Visualized traffic trends with matplotlib, providing actionable insights for network security monitoring.

SKILLS

- Programming Languages:** Java, Python, C, JavaScript
- Data Science and Machine Learning :** NumPy, Pandas, Scikit-learn(Sklearn), LangChain, Matplotlib, PyTorch
- Web Development::** HTML, CSS, React
- Database::** MySQL, MongoDB
- Tools & Technologies::** VS Code, Jupyter Notebook, Git/GitHub, Cisco Packet Tracer, Tinkercad, Google Colab

ACHIEVEMENTS

- Solved 400+ LeetCode Questions
- Achieved 5-Star Gold Badge in Problem Solving (HackerRank)